COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 6

2 Complexity Theory (tg508)

Time and space complexity

(*a*) Provide a precise definition of the following complexity classes: EXP, NEXP, and NPSPACE. [3 marks]

Answer: $\mathsf{EXP} = \bigcup_{k\in\mathbb{N}} \mathsf{DTIME}\left(2^{n^k}\right)$; i.e., the set of all decision problems that can be solved in time 2^{n^k}, for a integer k, by a deterministic Turing machine.

$\mathsf{NEXP} = \bigcup_{k\in\mathbb{N}} \mathsf{NDTIME}\left(2^{n^k}\right)$; i.e., the set of all decision problems that can be solved in time 2^{n^k}, for a integer k, by a non-deterministic Turing machine.

$\mathsf{NPSPACE} = \bigcup_{k\in\mathbb{N}} \mathsf{SPACE}(n^k)$; i.e., the set of all decision problems that can be solved in space n^k, for a integer k, by a non-deterministic Turing machine.

Space complexity

(*b*) Prove that $\mathsf{NPSPACE} \subseteq \mathsf{EXP} \subseteq \mathsf{NEXP}$. [7 marks]

Answer: The second inclusion follows immediately from the definitions. For the first inclusion, by Savitch's theorem $\mathsf{NPSPACE} = \mathsf{PSPACE}$. To show that $\mathsf{PSPACE} \subseteq \mathsf{EXP}$, let $S \in \mathsf{PSPACE}$. Then S can be decided via a TM with space complexity n^k, for some k. Hence the possible number of configurations is 2^{n^k}, and so an EXP-machine can enumerate over all configurations and emulate the PSPACE-machine.

Nondeterminism

(*c*) Prove that if $\mathsf{P} = \mathsf{NP}$, then $\mathsf{EXP} = \mathsf{NEXP}$. [10 marks]

Answer: Suppose that $\mathsf{P} = \mathsf{NP}$. The proof is by a padding argument. Let $S \in \mathsf{NEXP}$. Then the problem $S' = \{x01^{2^{|x|^k}} : x \in S\}$ is in NP, and in turn, by our assumption, $S' \in \mathsf{P}$. Hence, given $x \in S$, we can generate the input $x01^{2^{|x|^k}}$ in exponential time and solve it in polynomial time in the size of its input, hence still in exponential time, and so $S \in \mathsf{EXP}$.